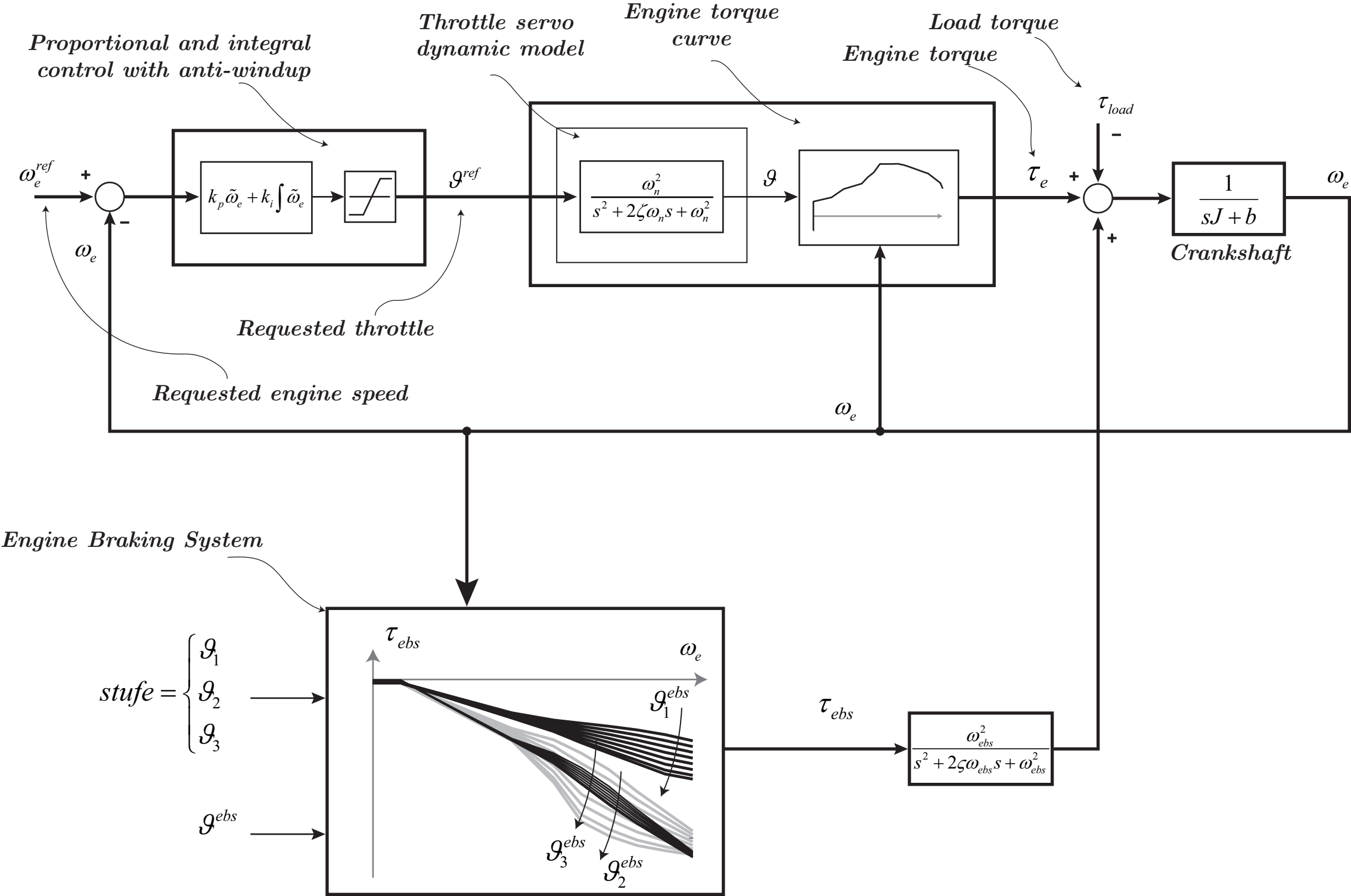
Proportional and integral control with anti-windup
Throttle servo dynamic model
Engine torque curve
Engine torque
Load torque
ω_e^{ref}
+
−
$k_p\tilde{\omega}_e + k_i\int\tilde{\omega}_e$
ϑ^{ref}
$\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$
ϑ
τ_e
τ_{load}
$\frac{1}{sJ+b}$
ω_e
Crankshaft
Requested throttle
Requested engine speed
Engine Braking System
$stufe = \begin{cases}\vartheta_1\\ \vartheta_2\\ \vartheta_3\end{cases}$
ϑ^{ebs}
τ_{ebs}
ϑ_1^{ebs}
ϑ_2^{ebs}
ϑ_3^{ebs}
$\frac{\omega_{ebs}^2}{s^2+2\zeta\omega_{ebs}s+\omega_{ebs}^2}$